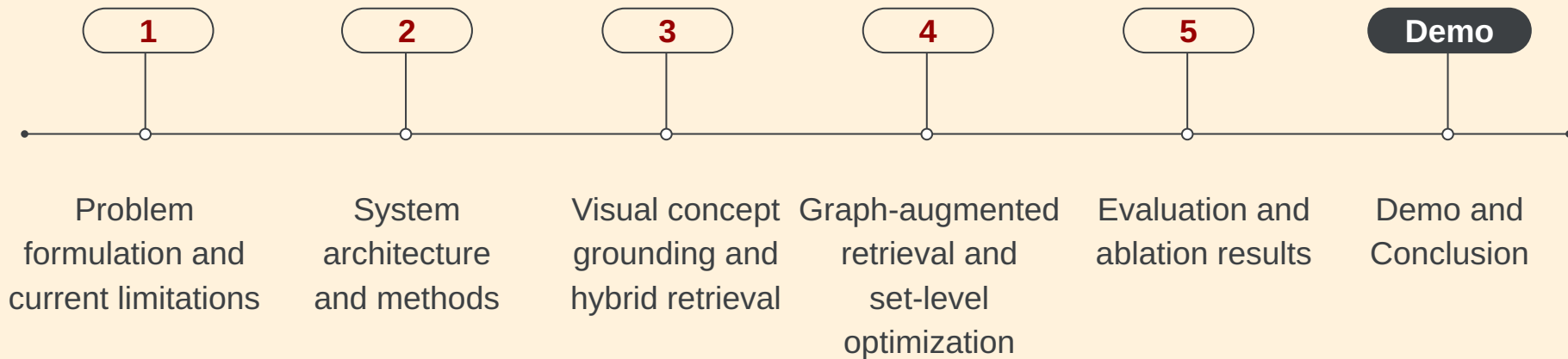


SmartMatch: A Multi-Agent Pipeline for Mood Board Generation via Graph-Augmented Retrieval and Multimodal Synthesis

Team 04

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Agenda



Cross-Modal Retrieval under Abstract Semantic Constraints

Task

- Map abstract natural language queries → *coherent image sets*

Limitation of existing methods

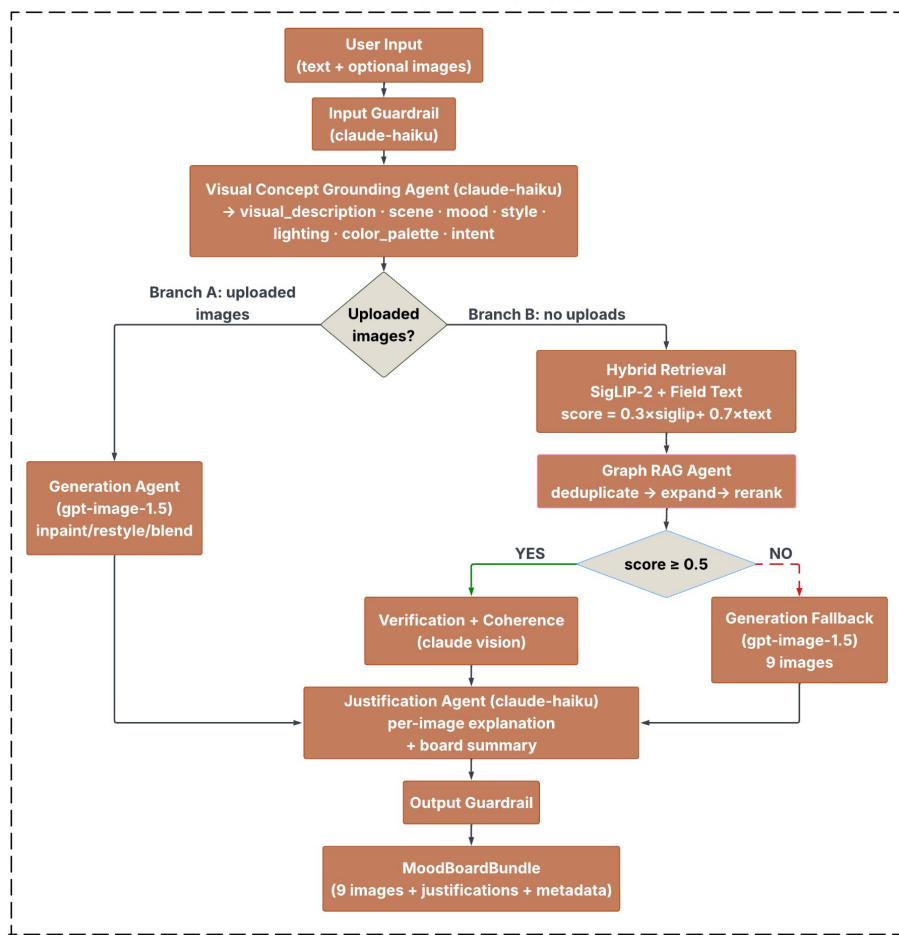
- Vision-language models optimize **pairwise embedding similarity**
- Assume alignment between textual and visual representations

Core challenge

- Abstract semantics (e.g., emotion, atmosphere) are **not explicitly grounded** in visual embedding space
- ⇒ leads to **systematic cross-modal misalignment**

Orchestrator Agent

(manages routing · step timing · logging · state)



Pipeline structure

- **Visual grounding** → structured query representation
- **Hybrid retrieval** → candidate image set (multi-signal similarity)
- **Graph RAG** → deduplication, neighbor expansion, reranking
- **Verification + coherence** → set-level selection
- **Generation fallback** → handle low-confidence

Structured Query Representation and Hybrid Retrieval

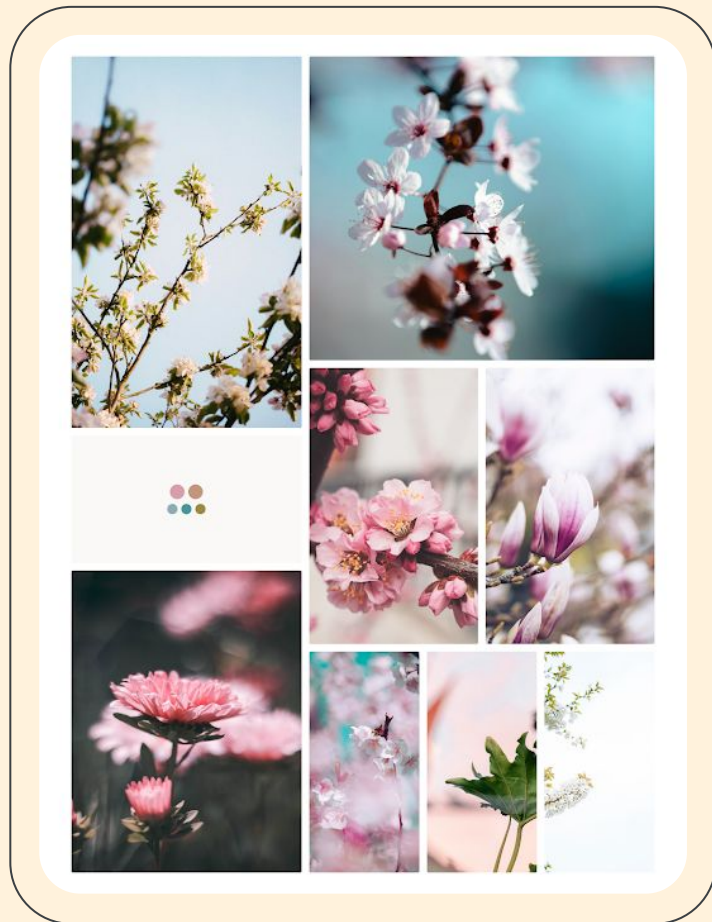
Visual concept grounding

- LLM transforms input text into structured fields: visual description · mood · color palette · lighting
- Acts as a **multi-field query rewriting mechanism**

Hybrid retrieval

$$\text{score} = 0.3 \cdot s_{\text{siglip}} + 0.7 \cdot s_{\text{text}}$$

- **SigLIP-2 embeddings** → global image–text similarity
- **Field-level text embeddings** → semantic alignment across attributes



Graph construction

- Build weighted graph over image corpus
- Edge weight captures visual and semantic similarity:

Weight = 0.35·s_mood + 0.20·s_color + 0.45·s_visual

Graph operations

- **Deduplication** → remove near-identical candidates
- **Neighbor expansion** → introduce diverse
- **Connectivity-based reranking** → promote well-integrated candidates

Key property

Transforms retrieval into a **set-level optimization problem**, balancing coherence and diversity through graph structure



Set-Level Selection and Adaptive Routing

Coherence-aware selection

- Multimodal verification evaluates alignment between images and query
- Selects final 9-image set balancing:
 - **thematic consistency**
 - **visual diversity**

Adaptive routing

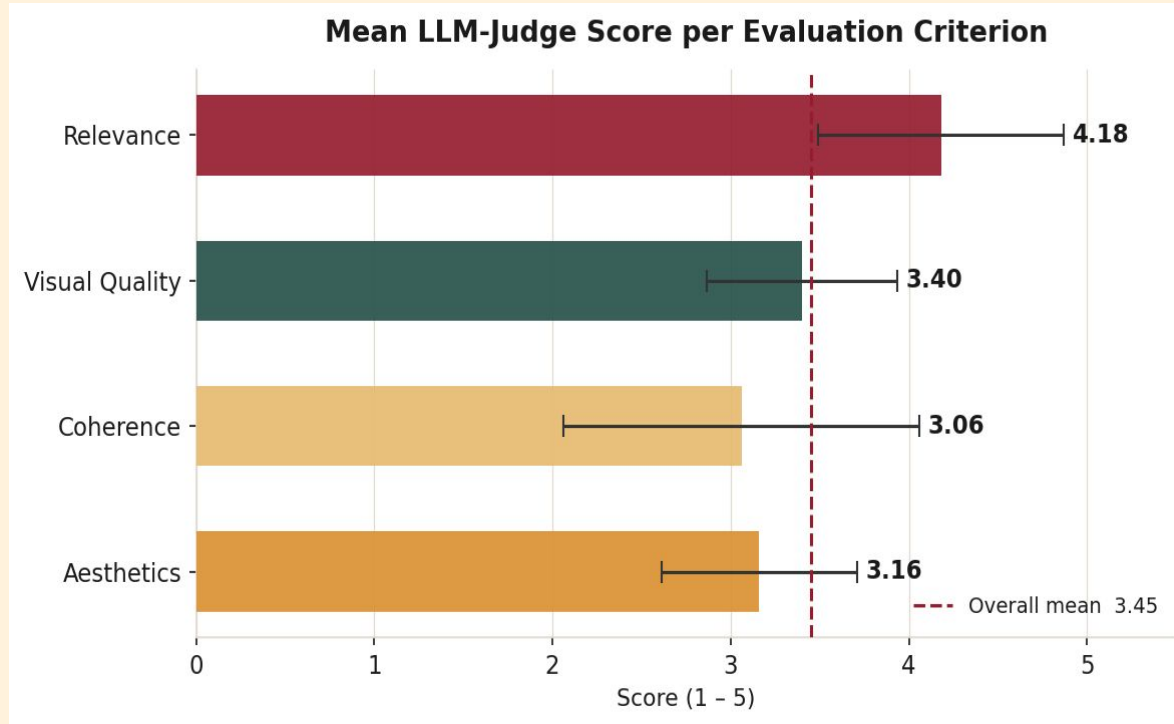
- If hybrid score < threshold → **generation fallback**
- Handles queries outside corpus distribution

Key property

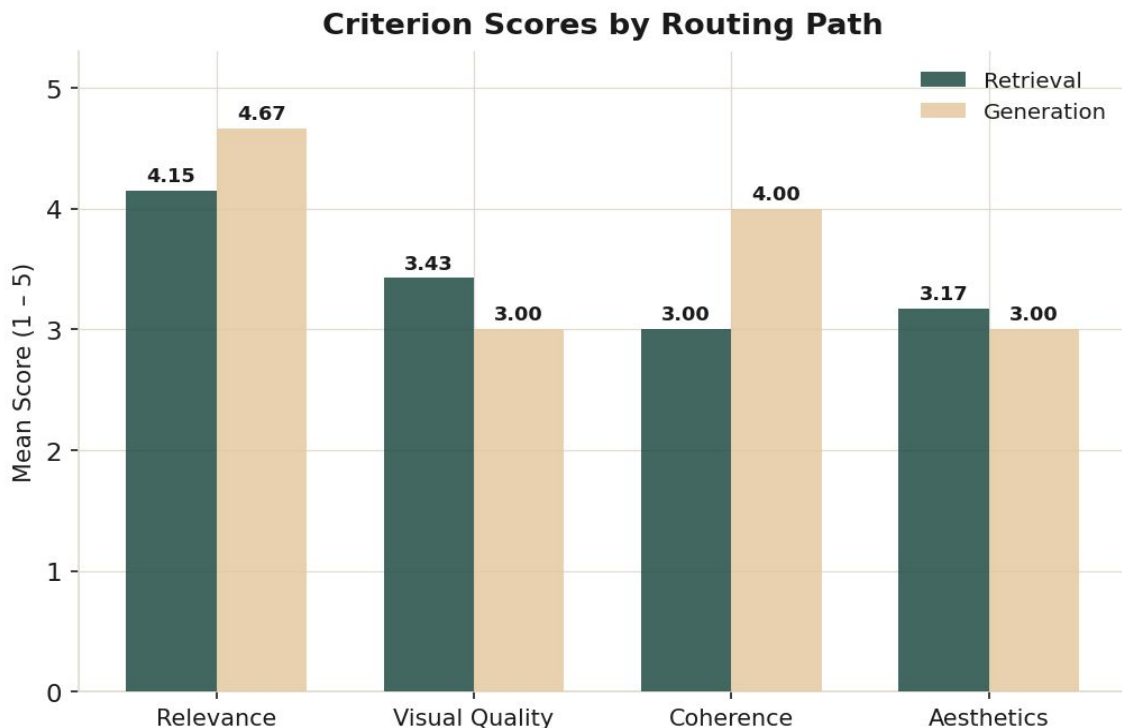
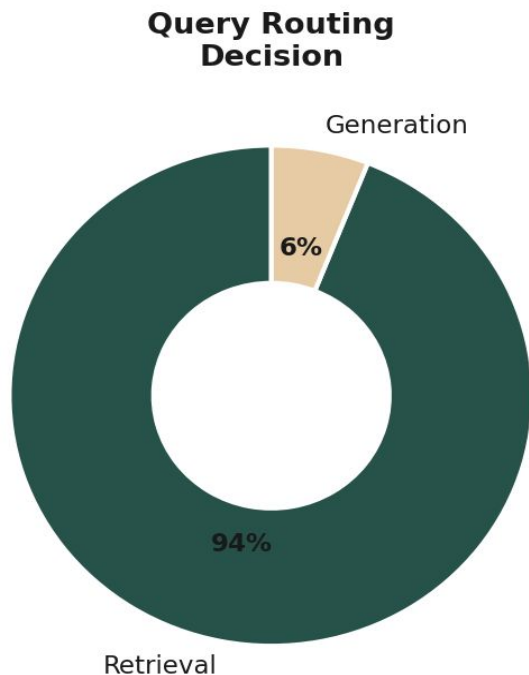
Final output is determined by **joint optimization over candidates**, not independent ranking

Evaluation and Ablation

04/24/2026



Mean LLM-judge scores across evaluation criteria (n = 50). Relevance is highest (4.18), while coherence is lowest (3.06), indicating that set-level consistency remains the primary limitation.



Query routing is dominated by retrieval (94%), yet generation achieves higher coherence (4.00 vs 3.00), indicating that retrieval-based outputs are limited by corpus homogeneity while generation improves diversity.

Conclusion

- **Grounding** resolves cross-modal misalignment
- **Hybrid retrieval + Graph RAG** enable coherence-aware selection
- **Generation fallback** improves diversity and coherence
- **Limitation:** Retrieval yields **visually homogeneous results**

Takeaway

Requires **joint modeling of representation alignment and set structure**,
not embedding similarity alone

Options

Upload your own images (optional)

Drag and drop files here

Limit 200MB per file • PNG, JPG, JPEG, WEBP

Browse files

Style reference image (optional)

Drag and drop file here

Limit 200MB per file • PNG, JPG, JPEG, WEBP

Browse files

SmartMatch · DSAN 6725 · Team 04

SmartMatch

Describe a feeling, moment, or idea — SmartMatch finds or generates the perfect mood board.

Describe what you're looking for

eclectic colorful apartment

Generate Board

Regenerate all

Board summary: # Mood Board Analysis: Eclectic Colorful Apartment

This collection captures the essence of intentional, joyful maximalism through a consistently vibrant yet warm color story—jewel tones, sunset yellows, and hot pinks unified by natural light that brings each hue to life. The images balance intimate, tactile details (paint-stained hands, ceramic collections, beaded textures) with expansive spatial views, suggesting a home where color permeates every surface and layer, from walls and windows to handmade objects and art supplies. Together, they tell the story of a lived-in, creative space that celebrates bold self-expression while maintaining visual harmony through consistent warm lighting and a fearless approach to mixing patterns, textures, and chromatic density.

Image 1

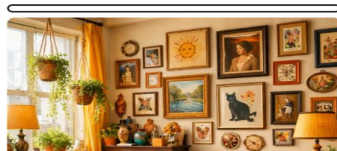
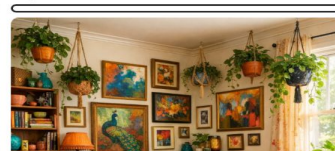


Image 2



Image 3



Demo

<https://huggingface.co/spaces/NandiniKodali/smartmatch>



Q&A

Thank you!